InvestFlex

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Background

"**Young investors** (aged 18–24) are **subject to** an ever-growing number of **investment scams**"

- Lloyds Bank 2024

"Although more and more **young people** begin to invest, they have **not been provided** with the **correct tools**"

- World Economic Forum 2024

Project Aim

The Customer - Young adults, typically only beginning to start their investing journey.

The Problem – The ethics standards of online investment platforms currently available which are seriously under question. With many facing lawsuits from various regulatory bodies or even complete bans (Bignell, F 2023 & Broverman, A 2023).

The Objective - Provide an analytical tool where users can research and learn about various companies and their stocks, in a reliable & ethical manner.

Technologies Used

Client-Side (Frontend)

React – A JavaScript library used to build user interfaces

HTML – Used for the structuring of the application

CSS – Used for the styling of the application

Key Packages:

Axios – Used to make requests to and from the server

React-Router – Used for routing/navigation within the app

Bootstrap – Used for UI development

Server-Side (Backend)

Ruby – Server-side programming language

Ruby on Rails – Ruby framework used to create API application

SQLite – Default Rails database to store information

Key Packages:

BCrypt – Used for password hashing of user's passwords

Rack-CORS – Resource sharing & frontend/backend interactions

SQLite3 – Local storage database

Design

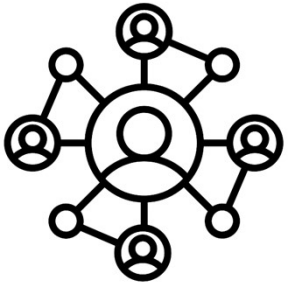
The application consists of **5** main sections:

Home

Blended Analysis

Portfolio

Learning



Community

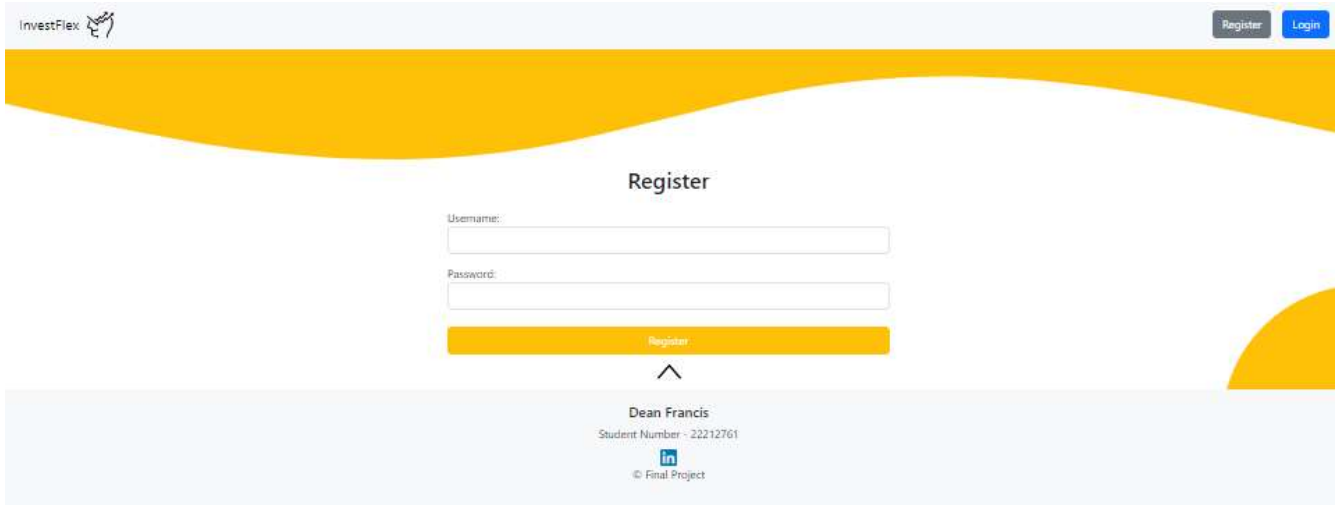
Demonstration

Frontend - Registration & Login Sections

Here users may register a username and password to access the full application.

Additionally, any user who has already registered with the application, may either log into their account or delete their account.

Validation measures have also been added such as unique usernames required.



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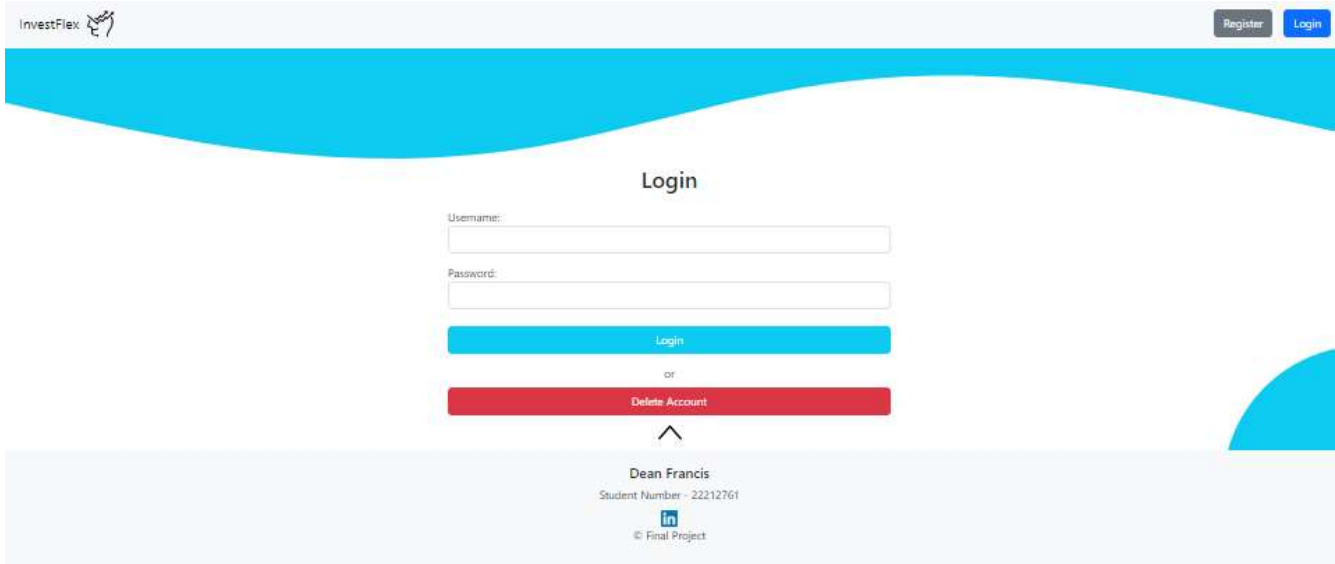
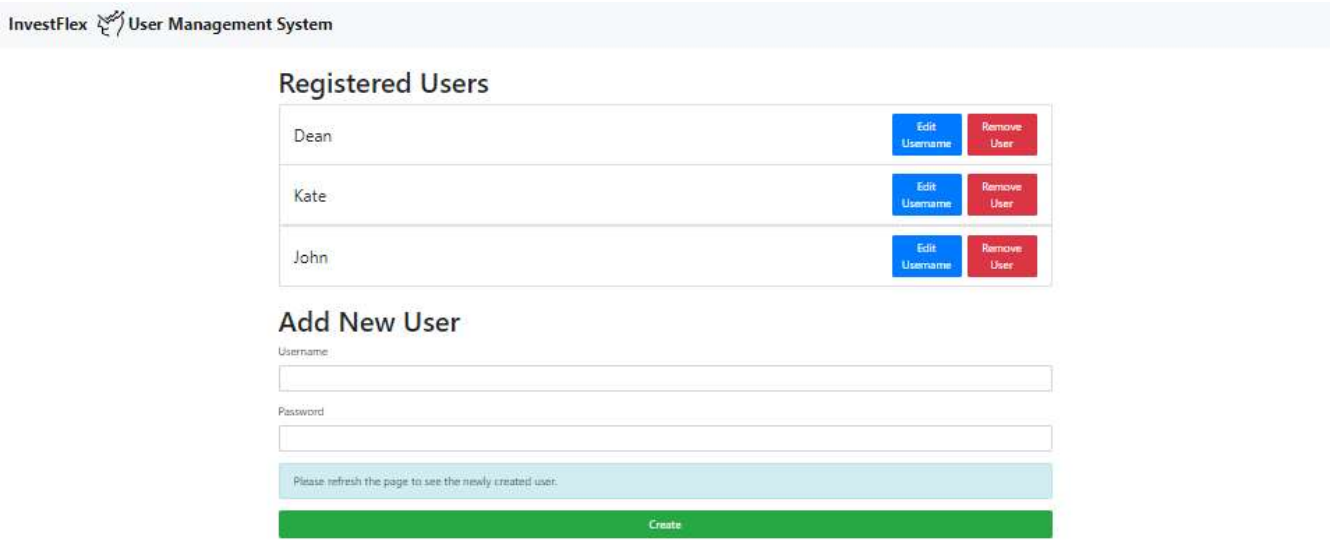


Demonstration

Backend – User Management System

Within the Ruby on Rails backend is the user management system.

Here administrators of the system can view all users currently registered and perform CRUD operations, such as editing usernames and deleting users.



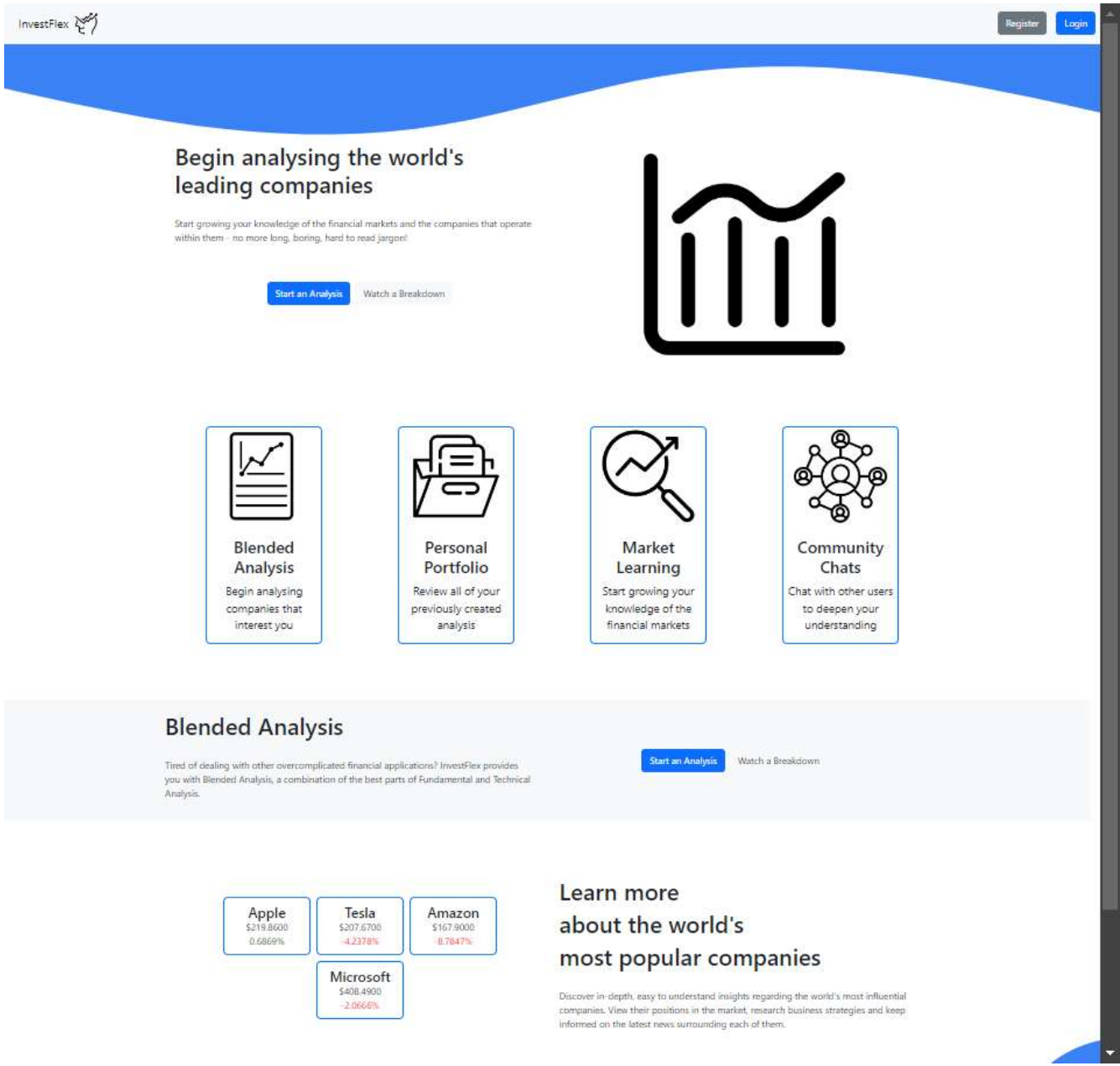
Demonstration

Frontend - Home Section

Here users are introduced to the application.

They may navigate through the various sections via the top navbar or the cards within the page itself.

They are also given some examples of companies they could analyse.



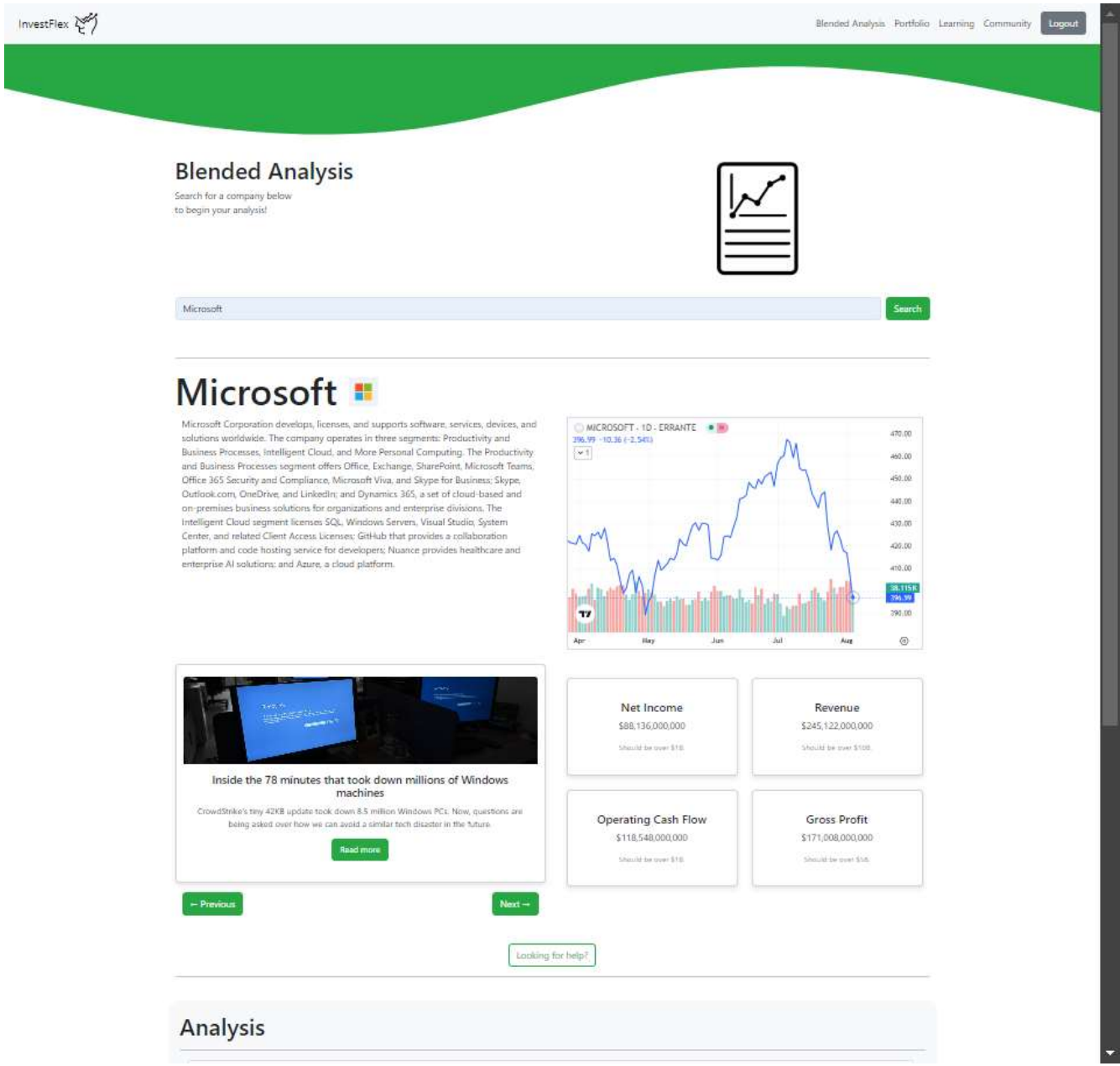
Demonstration

Blended Analysis Section

Here users may complete a blended analysis of a company.

It consists of 3 main sections:

- A search bar to find a company via its name or ticker symbol (Microsoft/MSFT).
- A research area comprising of the company's brief, price chart, news and metrics.
- An analysis area, where users input their findings and saved their analysis.

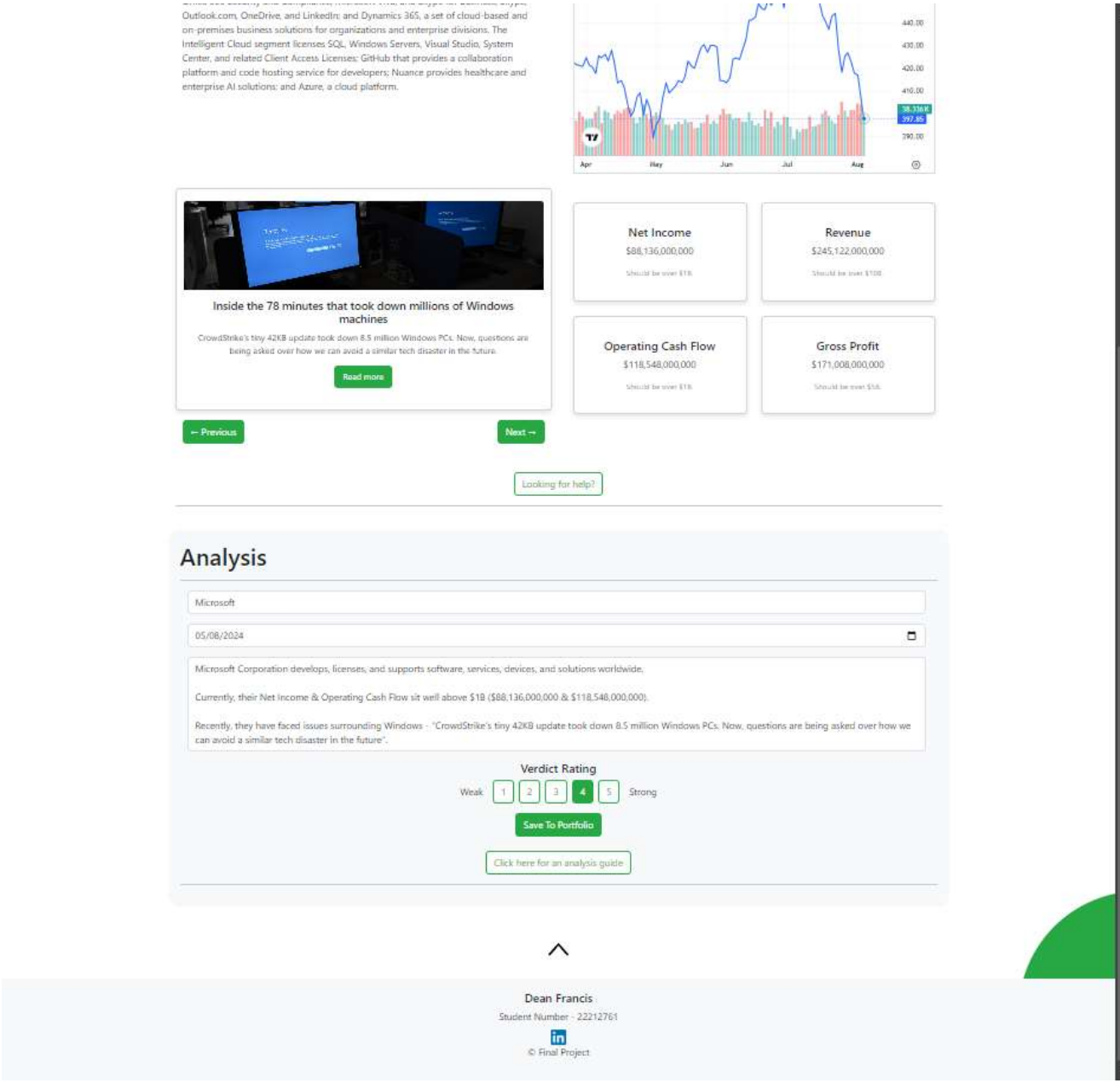


Demonstration

Analysis Section

Users should ask:

- Who is the company?
- What do they do?
- How are they performing?
- Why are they performing the way they are?



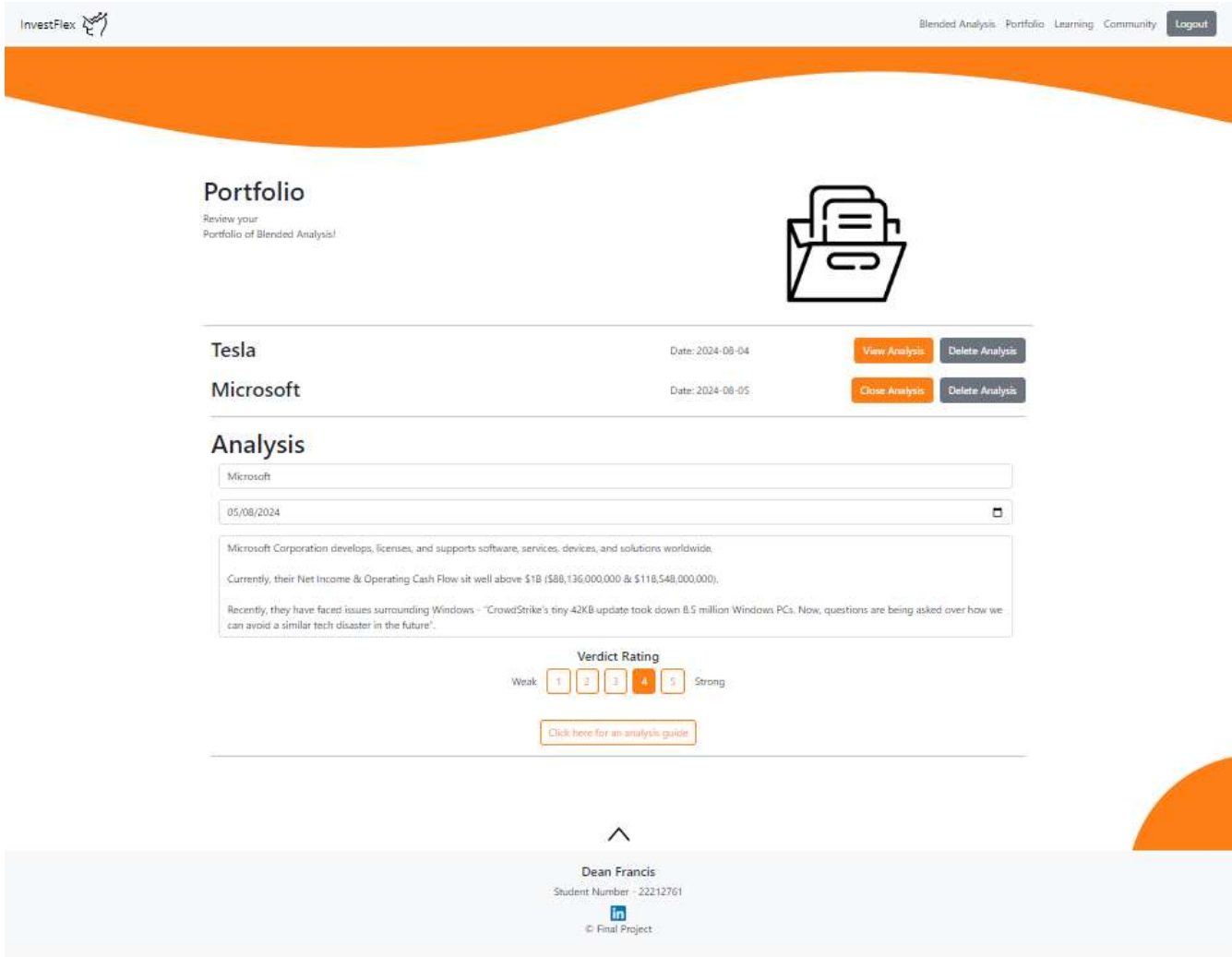
Demonstration

Portfolio Section

Here users can view their previously saved Blended Analysis for reflection or comparison to present performance.

The user may click “View Analysis” to open the full description.

The user may click “Delete Analysis” to remove it completely.



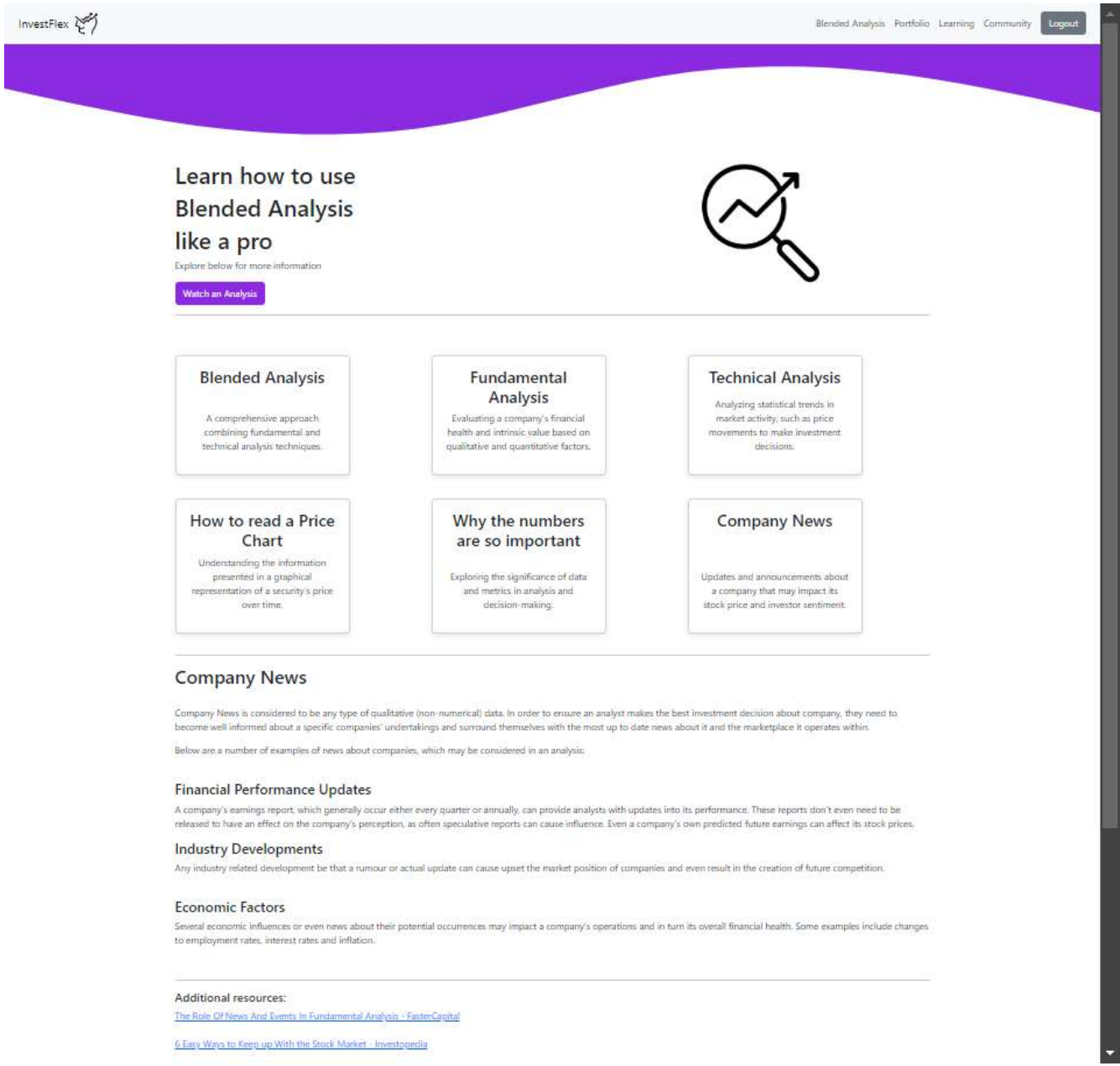
Demonstration

Learning Section

Here users can learn more about the InvestFlex application. All features found in the application are explained here.

It is broken down into 6 areas:

- Blended Analysis
- Fundamental Analysis
- Technical Analysis
- How to read a price chart
- Why the numbers are so important
- Company News



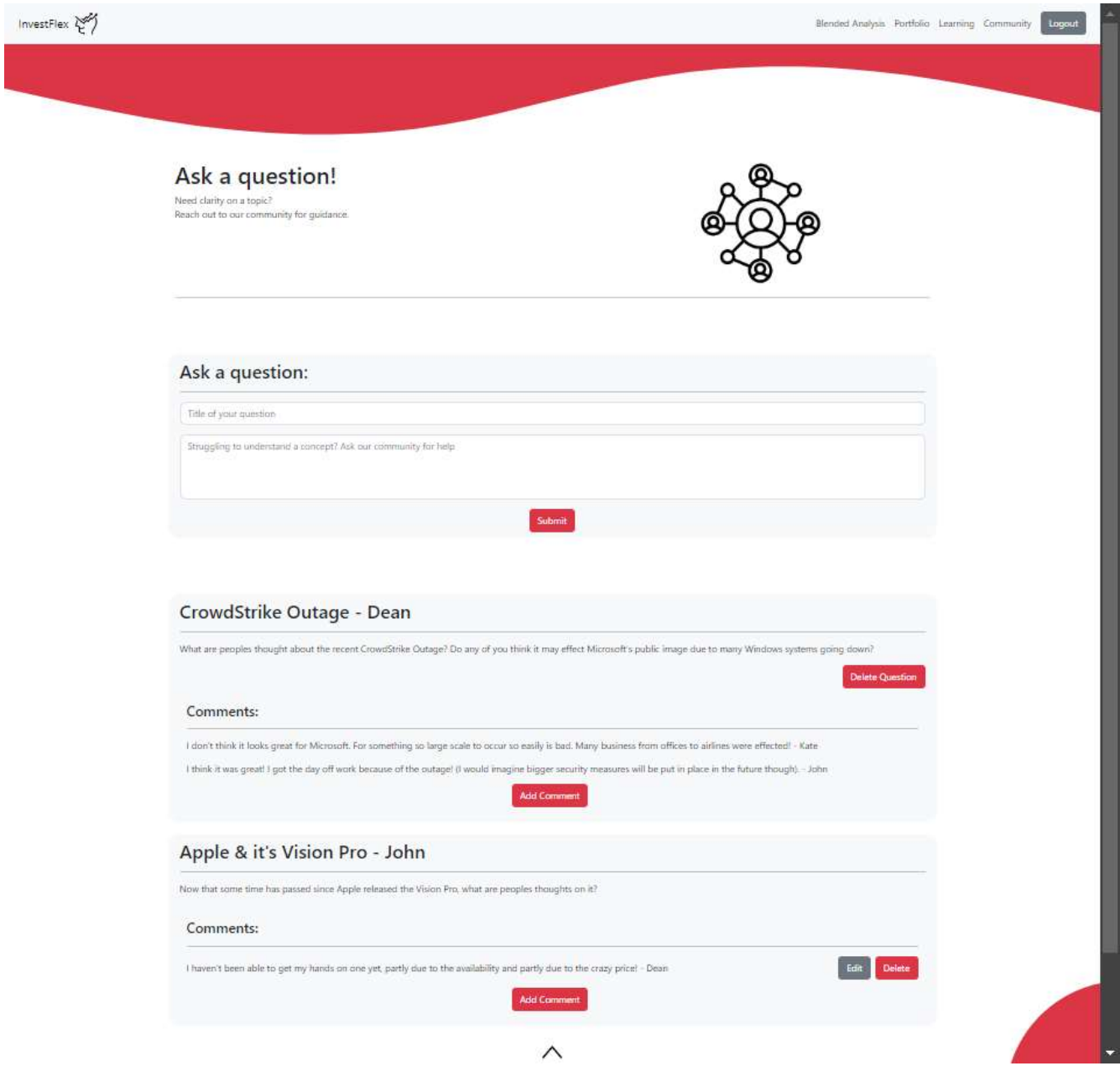
Demonstration

Community Section

Here users can ask questions about companies to further deepen their knowledge.

Any user may ask a question and submit a comment.

A user may only delete their respective question or edit/delete their respective comment.



Testing

The InvestFlex application contains both Unit Testing and Integration Testing throughout.

This is to ensure the applications individual components work as required, as well as, observing how the applications various parts interact with each other.

Unit Testing:

Comment Component Tests: Testing in isolation, adding a new comment/editing an existing comment/deleting an existing comment.

PortfolioList Component Tests: Testing in isolation, the deletion of a saved analysis item.

Integration Testing:

CommentsControllerTest: Testing the interaction between the CommentsControler and other components such as User and Questions.

UserControllerTest: Testing the interaction between the UserController and other components to verify user creation, login and valid credentials.

Evaluation

The evaluation of the application has been split into **2** sections, Testing and Success Conditions.

Testing:

- As mentioned in the previous slide, various testing methods have been incorporated throughout the application, to ensure it performs as expected.

Success Conditions:

Various conditions set out at the project's conception (Requirements Specification) which should be true.

- User should be able to register a profile.
- User should be able to log into a registered profile.
- Registered user should be able to access the all features of the application.
- Registered user should be able to search for a company, conduct a blended analysis and save it.

Advantages & Limitations

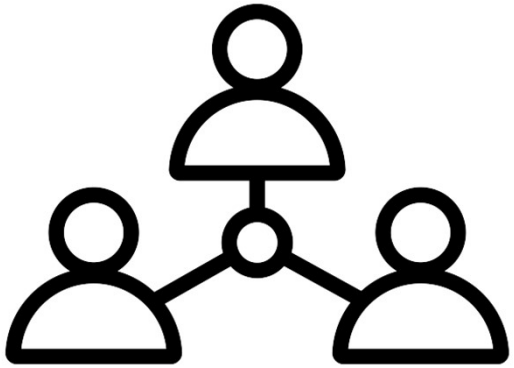
Current Limitations:

Searching for a company to analyse is subject to the external API's naming conventions, for example it must be the companies name used on the stock market - Facebook vs Meta

At present users cannot share their completed analysis in the community section.

Opportunities in the future:

The application could be incorporated into an already existing investment platform, to extend their reach to this projects target demographic – young investors (aged 18–24).

Through further development it could become a fully comprehension learning, research and investment experience.

Thank you for listening